

Minimal winding in the self-similar collapse of three unequal point vortices and of two concentric vortex polygons

Chase Hendrick

GENChase

Abstract. In a self-similar collapse of point vortices every vortex moves on a logarithmic spiral, and the dimensionless product $P = \omega_0 t_c$ of the initial angular velocity and the collapse time measures how tightly the spiral winds. We minimize P in closed form over two classical collapsing families. For three vortices with circulations $(1, 1/2, -1/3)$ and zero angular impulse, the collapsing configurations form two arcs, one for each orientation of the vortex triangle, and P has a unique critical point, a minimum, on each. The two minima, 1.0647059762... and 2.2038550160..., are the positive roots of the irreducible polynomial $8748x^6 - 49005x^4 + 27794x^2 + 18723$; they are given by the trigonometric form of Cardano's formula and are not expressible by real radicals. The smaller one is the minimum over the whole family. For equal circulations $\Gamma_1 = \Gamma_2$ the two arcs are equivalent and the minimum is $\sqrt{2}$. For two concentric regular n -gons carrying circulations x_n and -1 , $P = (K_n - \sqrt{2n-1} \cos n\theta) / (2n \sin n\theta)$ in terms of the relative rotation θ , with an explicit constant K_n , and its minimum is $\sqrt{K_n^2 - 2n + 1} / (2n)$; for pentagons this is $\sqrt{31682}/80$. The results are checked against the Biot–Savart velocities in 60-digit arithmetic and by direct integration of the equations of motion.

Keywords: point vortices; vortex collapse; self-similar motion; logarithmic spiral.

MSC 2020: 76B47, 37N10.

1. Introduction

Three point vortices can collapse to a point in finite time only if $1/\Gamma_1 + 1/\Gamma_2 + 1/\Gamma_3 = 0$ and their angular impulse vanishes, and under these conditions the motion is self-similar: the vortex triangle keeps its shape while it shrinks and rotates [1, 8]. Such collapse was found by Gröbli [6] and rediscovered by Novikov and Sedov [10], who also constructed collapsing configurations of four and five vortices. Aref [3] gave the collapse rate and the rotation frequency for general circulations, and Gotoda [5] gave an explicit parametrization of the collapsing configurations. Collapse also occurs in configurations of higher symmetry. Aref [2] reduced the motion of two concentric regular n -gons of vortices to an integrable Hamiltonian system with two degrees of freedom, and Koiller et al. [7] found collapsing two-ring configurations, whose vortices move on logarithmic spirals.

In a self-similar collapse at time t_c the configuration rotates through the angle $\omega_0 t_c \ln(1 - t/t_c)^{-1}$ by time t , where ω_0 is the initial angular velocity. Each vortex therefore moves on a logarithmic spiral about the collision point, and the dimensionless number $P = |\omega_0| t_c$, which is invariant under rescaling of lengths, times and circulations, is the angle through which the configuration turns while the square of its size decreases by the factor e . Equivalently, the path of each vortex makes the constant angle $\arctan 2P$ with the direction to the collision point, and a vortex that starts at distance r_0 from that point travels the distance $r_0 \sqrt{1 + 4P^2}$ before the collapse. Within a collapsing family it is natural to ask which configuration winds least.

We answer this question for two families. In Section 3 we treat three vortices with circulations $(1, 1/2, -1/3)$. The collapsing configurations form two arcs, one for each orientation of the triangle,

and P has a unique minimum on each; the two minima are distinct conjugate algebraic numbers of degree six, and the smaller is the minimum over the whole family. For $\Gamma_1 = \Gamma_2$ the two orientations are exchanged by interchanging the equal vortices and the minimum is $\sqrt{2}$ (Remark 3). In Section 4 we treat two concentric regular n -gons, for which the minimization reduces to an elementary inequality once P is written in a suitable form. Section 5 describes an independent numerical check.

2. Self-similar collapse

The positions $z_j \in \mathbb{C}$ of point vortices with circulations Γ_j evolve according to

$$\dot{\bar{z}}_j = \frac{1}{2\pi i} \sum_{k \neq j} \frac{\Gamma_k}{z_j - z_k}. \quad (1)$$

Suppose $\sum_j \Gamma_j \neq 0$, let $z_c = \sum_j \Gamma_j z_j / \sum_j \Gamma_j$ be the center of vorticity, which is conserved, and suppose that at $t = 0$ there is $\kappa \in \mathbb{C}$ with $\dot{z}_j = \kappa(z_j - z_c)$ for all j . The right-hand side of (1) is homogeneous of degree -1 in the relative positions and commutes with rotations, so the solution is $z_j(t) = z_c + \lambda(t)e^{i\varphi(t)}(z_j(0) - z_c)$ with $\lambda\dot{\lambda} = \operatorname{Re} \kappa$ and $\lambda^2\dot{\varphi} = \operatorname{Im} \kappa$. Hence $\lambda^2 = 1 + 2\operatorname{Re} \kappa t$. If $\operatorname{Re} \kappa < 0$ the vortices collide at z_c at the time $t_c = -1/(2\operatorname{Re} \kappa)$, and $\omega_0 = \operatorname{Im} \kappa$, so that

$$P = |\omega_0| t_c = \frac{|\operatorname{Im} \kappa|}{-2\operatorname{Re} \kappa}. \quad (2)$$

Integrating $\dot{\varphi} = \omega_0/\lambda^2$ gives $\varphi = -P \ln \lambda^2$ when $\omega_0 > 0$, which is the description of the spiral given in the introduction. The angular impulse about the center of vorticity satisfies

$$\sum_j \Gamma_j |z_j - z_c|^2 = \frac{1}{\sum_j \Gamma_j} \sum_{j < k} \Gamma_j \Gamma_k |z_j - z_k|^2. \quad (3)$$

It is conserved and, in a self-similar motion, proportional to λ^2 , so it must vanish for a collapse. The minimizations below reduce to the following elementary inequality.

Lemma 1. *Let $a > |b|$. For $0 < \alpha < \pi$, $(a - b \cos \alpha) / \sin \alpha \geq \sqrt{a^2 - b^2}$, with equality if and only if $\cos \alpha = b/a$.*

Proof. The numerator is positive, and $(a - b \cos \alpha)^2 - (a^2 - b^2) \sin^2 \alpha = (a \cos \alpha - b)^2$. $\square$

3. Three vortices with circulations $(1, 1/2, -1/3)$

Let $\Gamma = (1, 1/2, -1/3)$, so that $1/\Gamma_1 + 1/\Gamma_2 + 1/\Gamma_3 = 0$ and $\sum_j \Gamma_j = 7/6$. For $w = (z_3 - z_1)/(z_2 - z_1)$ the sum $\sum_{j < k} \Gamma_j \Gamma_k |z_j - z_k|^2$ in (3) equals $|z_2 - z_1|^2 (1/2 - |w|^2/3 - |w - 1|^2/6)$, which vanishes exactly on the circle $|w - 1/3| = \sqrt{7}/3$. We use the parametrization of the zero-impulse configurations given by Gotoda [5, Sect. 3], which for these circulations reads

$$z_1 = \frac{2}{9} \left(1 + \frac{\sqrt{7}}{2} e^{-i\theta} \right), \quad z_2 = \frac{2}{9} (1 - \sqrt{7} e^{-i\theta}), \quad z_3 = 1, \quad \theta \in [0, 2\pi). \quad (4)$$

Here $\sum_j \Gamma_j z_j = 0$, so $z_c = 0$, and

$$z_1 - z_2 = \frac{\sqrt{7}}{3} e^{-i\theta}, \quad z_3 - z_1 = \frac{\sqrt{7}}{9} (\sqrt{7} - e^{-i\theta}), \quad z_3 - z_2 = \frac{\sqrt{7}}{9} (\sqrt{7} + 2e^{-i\theta}). \quad (5)$$

The shape ratio is $w = 1/3 - (\sqrt{7}/3)e^{i\theta}$, which traverses the zero-impulse circle exactly once. Every zero-impulse configuration is therefore obtained, up to translation, rotation and dilation, for exactly one θ . Since $\text{Im } w = -(\sqrt{7}/3)\sin\theta$, the triangle $z_1 z_2 z_3$ is positively oriented for $\sin\theta < 0$ and negatively oriented for $\sin\theta > 0$.

Lemma 2. *For every θ the three quotients $\dot{z}_j/(z_j - z_c)$ are equal to*

$$\kappa = \frac{27i}{4\sqrt{7}\pi} \cdot \frac{\sqrt{7} + e^{i\theta}}{(\sqrt{7} - e^{i\theta})(\sqrt{7} + 2e^{i\theta})}. \quad (6)$$

Consequently, with $D(\theta) = (4 - \sqrt{7}\cos\theta)(11 + 4\sqrt{7}\cos\theta) > 0$,

$$\text{Re } \kappa = -\frac{27(14\cos\theta + \sqrt{7})\sin\theta}{28\pi D(\theta)}, \quad \text{Im } \kappa = \frac{27(14\sin^2\theta + 6\sqrt{7}\cos\theta + 21)}{28\pi D(\theta)}. \quad (7)$$

Proof. Since $z_c = 0$ and $z_3 = 1$, $\kappa = \dot{z}_3$, and by (1), $\overline{\dot{z}_3} = (2\pi i)^{-1}((z_3 - z_1)^{-1} + (1/2)(z_3 - z_2)^{-1})$. Inserting the differences above gives the stated κ , and the same computation for $j = 1, 2$ gives the same value. Multiplying numerator and denominator by $(\sqrt{7} - e^{-i\theta})(\sqrt{7} + 2e^{-i\theta})$ and using $|\sqrt{7} - e^{i\theta}|^2 = 2(4 - \sqrt{7}\cos\theta)$ and $|\sqrt{7} + 2e^{i\theta}|^2 = 11 + 4\sqrt{7}\cos\theta$ gives the real and imaginary parts. Both factors of D are positive because $\sqrt{7} < 4$ and $4\sqrt{7} < 11$. $\square$

The numerator of $\text{Im } \kappa$ is at least $21 - 6\sqrt{7} > 0$, so every zero-impulse configuration rotates in the positive sense, and by Lemma 2 it collapses exactly when $(14\cos\theta + \sqrt{7})\sin\theta > 0$. Let $\theta_0 \in (\pi/2, \pi)$ be defined by $\cos\theta_0 = -\sqrt{7}/14$. The collapsing configurations are those with θ in one of the two arcs

$$\mathcal{B} = (0, \theta_0), \quad \mathcal{A} = (\pi, 2\pi - \theta_0), \quad (8)$$

the triangle being negatively oriented on $\mathcal{B}$ and positively oriented on $\mathcal{A}$. By (2),

$$P(\theta) = \frac{14\sin^2\theta + 6\sqrt{7}\cos\theta + 21}{2(14\cos\theta + \sqrt{7})\sin\theta}, \quad \theta \in \mathcal{A} \cup \mathcal{B}. \quad (9)$$

Complex conjugation maps solutions of (1) to solutions with time reversed, and $z_j(-\theta) = \overline{z_j(\theta)}$, so $\theta \rightarrow -\theta$ maps $\mathcal{A}$ onto the expanding arc (θ_0, π) and $\mathcal{B}$ onto the expanding arc $(2\pi - \theta_0, 2\pi)$. No such symmetry relates $\mathcal{A}$ to $\mathcal{B}$.

Proposition 1. *The function P has exactly one critical point on each of $\mathcal{A}$ and $\mathcal{B}$, and it is the minimum of P on that arc. Let*

$$X = \frac{245351\sqrt{5201}}{5201^2}, \quad R = \frac{7\sqrt{5201}}{162}. \quad (10)$$

Then

$$\min_{\mathcal{A}} P = \sqrt{\frac{605}{324} + R \cos\left(\frac{1}{3} \arccos X - \frac{2\pi}{3}\right)} = 1.0647059762712043..., \quad (11)$$

$$\min_{\mathcal{B}} P = \sqrt{\frac{605}{324} + R \cos\left(\frac{1}{3} \arccos X\right)} = 2.2038550160361327..., \quad (12)$$

attained at $\cos \theta = -0.9243893679\dots$ and $\cos \theta = 0.6739838839\dots$ respectively. Both minima are roots of the polynomial $8748x^6 - 49005x^4 + 27794x^2 + 18723$, which is irreducible over $\mathbb{Q}$ and over $\mathbb{Q}(\sqrt{7})$ and whose real roots are exactly these two numbers and their negatives. In particular, the least value of P over all collapsing configurations is $1.0647059762712043\dots$

Proof. Write $c = \cos \theta$, so that $P = N(c)/(2(14c + \sqrt{7}) \sin \theta)$ with $N(c) = 35 + 6\sqrt{7}c - 14c^2$, which is positive on both arcs. Since $P > 0$ there and $dc/d\theta = -\sin \theta \neq 0$, the critical points of P on each arc correspond to the zeros of $d(P^2)/dc$ in the range of c on that arc. Now $P^2 = N^2/(4(14c + \sqrt{7})^2(1 - c^2))$, and a direct computation gives

$$\frac{d(P^2)}{dc} = \frac{\sqrt{7}N(c)C(c)}{4(1 - c^2)^2(14c + \sqrt{7})^3}, \quad C(c) = 196c^3 + 224\sqrt{7}c^2 + 14c - 128\sqrt{7}. \quad (13)$$

With $c = \sqrt{7}u$ we have $C = \sqrt{7}f(u)$, $f(u) = 1372u^3 + 1568u^2 + 14u - 128$. Now $f(-2) < 0 < f(-1)$, $f(-1/\sqrt{7}) = 96 - 30\sqrt{7} > 0$, $f(-1/14) = -243/2 < 0$ and $f(1/\sqrt{7}) = 96 + 30\sqrt{7} > 0$, so the cubic f has one root in each of $(-2, -1)$, $(-1/\sqrt{7}, -1/14)$ and $(-1/14, 1/\sqrt{7})$. The first gives $|c| > 1$. The second and third lie in the ranges of u on $\mathcal{A}$ and on $\mathcal{B}$, which are $(-1/\sqrt{7}, -1/14)$ and $(-1/14, 1/\sqrt{7})$. Hence each arc contains exactly one critical point. At the endpoints of both arcs the numerator of (9) takes the values $21 + 6\sqrt{7}$, $63/2$ and $21 - 6\sqrt{7}$, all positive, while the denominator vanishes; since $P > 0$ on the arcs, $P \rightarrow +\infty$ at every endpoint. A continuous function on an open interval that tends to $+\infty$ at both ends and has a single critical point attains its minimum there.

To find the minimal values we eliminate c between $C(c) = 0$ and $4P^2(14c + \sqrt{7})^2(1 - c^2) = N(c)^2$, and then eliminate $\sqrt{7}$. The resultant is a constant multiple of $(8748P^6 - 49005P^4 + 27794P^2 + 18723)^2$. As a cubic in $q = P^2$ this polynomial is irreducible over $\mathbb{Q}$ and has positive discriminant, so it has three real roots, $q \approx 4.856977$, 1.133599 and -0.388724 , given by $q_k = 605/324 + R \cos(1/3 \arccos X - 2\pi k/3)$, $k = 0, 1, 2$. The minima are positive roots of the sextic, so their squares are q_0 and q_1 , and evaluating P at the two critical points identifies q_0 with $\mathcal{B}$ and q_1 with $\mathcal{A}$. $\square$

Remark 1. The two minima are conjugate algebraic numbers of degree six. Since the cubic in q is irreducible over $\mathbb{Q}$ with three real roots, none of its roots is expressible by real radicals (casus irreducibilis), and hence neither is the square root of one.

Remark 2. The same construction applies to $\Gamma = (1, \mu, -\mu/(1 + \mu))$. Multiplying the circulations by $1/\mu$ and interchanging the first two vortices shows that the pair of branch minima is invariant under $\mu \rightarrow 1/\mu$. Numerically the two branch minima are $(1.3399, 1.4983)$ at $\mu = 0.9$, $(1.4067, 1.4218)$ at $\mu = 0.99$ and $(1.41346, 1.41496)$ at $\mu = 0.999$, and they tend to $\sqrt{2}$ as $\mu \rightarrow 1$.

Remark 3. For $\Gamma = (1, 1, -1/2)$ the zero-impulse circle is $|w - 1/2| = \sqrt{3}/2$. With $w = 1/2 + (\sqrt{3}/2)e^{i\varphi}$, $z_1 = 0$ and $z_2 = 1$, the same computation gives

$$\operatorname{Re} \kappa = \frac{3 \sin 2\varphi}{2\pi(3 \cos 2\varphi - 5)}, \quad \operatorname{Im} \kappa = \frac{3(\cos 2\varphi - 3)}{2\pi(3 \cos 2\varphi - 5)}, \quad P = \frac{3 - \cos 2\varphi}{2 \sin 2\varphi}, \quad (14)$$

on the collapsing arcs $0 < \varphi < \pi/2$ and $\pi < \varphi < 3\pi/2$, which are exchanged by interchanging the two equal vortices ($w \rightarrow 1 - w$). By Lemma 1, $P \geq \sqrt{2}$, with equality if and only if $\cos 2\varphi = 1/3$. This formula is a specialization and reparametrization of Gröbli's spiral coefficient [6, Sect. 10].

4. Two concentric regular polygons

Let $n \geq 2$ and $\varepsilon = e^{2\pi i/n}$, and place vortices of circulation $x > 0$ at $z\varepsilon^k$ and vortices of circulation -1 at $\zeta\varepsilon^k$, $k = 0, \dots, n-1$. The motion preserves this symmetry, the center of vorticity is the origin, and, since $\sum_{k=1}^{n-1} (1 - \varepsilon^k)^{-1} = (n-1)/2$ and $\sum_{k=0}^{n-1} (z - \zeta\varepsilon^k)^{-1} = nz^{n-1}/(z^n - \zeta^n)$, (1) reduces to [2, Eq. (3)]

$$\bar{z} = \frac{1}{2\pi i} \left(\frac{x(n-1)}{2z} - \frac{nz^{n-1}}{z^n - \zeta^n} \right), \quad \bar{\zeta} = \frac{1}{2\pi i} \left(-\frac{n-1}{2\zeta} + \frac{xn\zeta^{n-1}}{\zeta^n - z^n} \right). \quad (15)$$

A collapse requires the angular impulse $n(x|z|^2 - |\zeta|^2)$ to vanish, and after a rotation and a dilation we take $z = 1$ and $\zeta = \sqrt{x}e^{i\theta}$. The motion is self-similar exactly when $\dot{z}/z = \dot{\zeta}/\zeta$. Writing $w = \zeta^n$, (15) gives

$$\bar{\dot{z}}/z = \frac{1}{2\pi i} \left(\frac{x(n-1)}{2} - \frac{n}{1-w} \right), \quad \bar{\dot{\zeta}}/\zeta = \frac{1}{2\pi i} \left(-\frac{n-1}{2x} - \frac{nw}{1-w} \right), \quad (16)$$

and these are equal if and only if

$$(n-1)x^2 - 2nx + (n-1) = 0, \quad (17)$$

a condition that does not involve θ ; it is the circulation condition of Koiller et al. [7, Sect. 11]. Its roots are x_n and $1/x_n$, where

$$x_n = \frac{n + \sqrt{2n-1}}{n-1} = e^a, \quad \cosh a = \frac{n}{n-1}. \quad (18)$$

The root $1/x_n$ gives the same family with the two rings interchanged, as one sees by multiplying the circulations by $-x_n$ and reflecting the configuration, so we take $x = x_n$. Then every θ gives a self-similar motion. Let $\rho = x_n^{n/2} = e^{na/2} > 1$ and $\alpha = n\theta$, so that $w = \rho e^{i\alpha}$. The common value of $\bar{\kappa}$ is $S/(2\pi i)$ with $S = x_n(n-1)/2 - n/(1-w)$, so $\kappa = (\text{Im } S + i \text{Re } S)/(2\pi)$, and since $\text{Im } (1-w)^{-1} = \rho \sin \alpha / |1-w|^2$,

$$\text{Re } \kappa = -\frac{n\rho \sin \alpha}{2\pi |1-w|^2}. \quad (19)$$

The configuration therefore collapses exactly when $\sin n\theta > 0$, that is, for $0 < \theta < \pi/n$ modulo $2\pi/n$. Using $|1-w|^2 = 1 - 2\rho \cos \alpha + \rho^2$ one finds

$$\frac{|1-w|^2 \text{Re } S}{\rho} = K_n - ((n-1)x_n - n) \cos \alpha, \quad K_n = \frac{(n-1)x_n(\rho + \rho^{-1})}{2} - \frac{n}{\rho}. \quad (20)$$

Here $(n-1)x_n - n = \sqrt{2n-1}$, and substituting $n = (n-1) \cosh a$ in n/ρ gives $K_n = (n-1) \sinh((n+2)a/2)$. Moreover

$$K_n - \sqrt{2n-1} = \frac{(n-1)x_n(\rho^{1/2} - \rho^{-1/2})^2}{2} + n(1 - \rho^{-1}) > 0, \quad (21)$$

so $\text{Re } S > 0$, and (2) gives

$$P = \frac{K_n - \sqrt{2n-1} \cos n\theta}{2n \sin n\theta}, \quad 0 < n\theta < \pi. \quad (22)$$

The collapse rates as functions of the relative angle are given by Koiller et al. [7, Sect. 11]; (22) expresses their ratio with the constant K_n explicit.

Proposition 2. *For $n \geq 2$ the collapsing configurations of two concentric regular n -gons with circulations x_n and -1 satisfy*

$$P \geq F_n = \frac{\sqrt{K_n^2 - (2n-1)}}{2n}, \quad (23)$$

with equality if and only if $\cos n\theta = \sqrt{2n-1}/K_n$.

Proof. Apply Lemma 1 to (22) with $a = K_n$ and $b = \sqrt{2n-1}$. □

For small n the constants are as follows.

n	x_n	K_n	F_n	$\cos n\theta$ at the minimum
2	$2 + \sqrt{3}$	$4\sqrt{3}$	$3\sqrt{5}/4$	$1/4$
3	$(3 + \sqrt{5})/2$	11	$\sqrt{29}/3$	$\sqrt{5}/11$
4	$(4 + \sqrt{7})/3$	$55\sqrt{7}/9$	$\sqrt{322}/9$	$9/55$
5	2	$127\sqrt{2}/8$	$\sqrt{31682}/80$	$12\sqrt{2}/127$

For $n = 2$ the configuration is a parallelogram of four vortices, a case treated by Novikov and Sedov [10, Sect. 4]. For $n = 5$ the circulation ratio is rational, $x_5 = 2$, and

$$P = \frac{127\sqrt{2} - 24 \cos 5\theta}{80 \sin 5\theta} \geq \frac{\sqrt{31682}}{80} = 2.2249297741726591.... \quad (24)$$

5. Numerical verification

The results were checked by a computation that does not use the formulas above. It evaluates the Biot–Savart velocities (1) directly, tests self-similarity by comparing $\dot{z}_j/(z_j - z_c)$ across all vortices, and obtains P from (2). For three vortices the zero-impulse configurations were generated independently of (4), by fixing two vortices and moving the third around the zero-impulse circle. The computations used 60 significant digits (mpmath) unless stated otherwise, and the algebraic steps in the proofs, including Lemma 2, the factorization of $d(P^2)/dc$, the resultant and the irreducibility statements, were verified in exact arithmetic (SymPy).

Check	Result
Three vortices: spread of the three quotients $\dot{z}_j/(z_j - z_c)$ relative to $ \kappa $, 4000 shapes on the zero-impulse circle	$\leq 9 \times 10^{-61}$; 2000 of the 4000 shapes collapse
Local minima of P over the collapsing shapes, refined by solving $dP/d\varphi = 0$ numerically	two; they differ from the values in Proposition 1 by $\leq 2 \times 10^{-61}$
Formula (9) against direct evaluation at the positions (4), 399 interior points of each arc	relative difference $\leq 10^{-59}$
Integration of (1) from the minimizing configuration on $\mathcal{A}$ (Taylor method, 30 digits)	$ z_j - z_c ^2$ follows $1 - t/t_c$ to relative accuracy 5×10^{-31} up to $t = 0.9 t_c$
Two rings, $n = 2, \dots, 8$: P from (1) against (22) over the collapsing range, and its numerical minimum against F_n	relative difference $\leq 3 \times 10^{-58}$; minima agree to 2×10^{-60}

6. Discussion

The expression (9) is a special case of the rates of Gröbli [6] and Aref [3], and the content of Proposition 1 is its minimization over the whole collapsing family. What distinguishes unequal circulations is that the two orientations of the triangle are no longer equivalent. For $\Gamma_1 = \Gamma_2$ the interchange of the equal vortices maps one collapsing arc onto the other and the two branch minima coincide (Remark 3); for $\Gamma_1 \neq \Gamma_2$ no symmetry relates the arcs, the minima separate (Remark 2), and a minimization restricted to one orientation of the triangle finds only one of them.

Leoncini, Kuznetsov and Zaslavsky [9] analyze the motion near collapse when two of the vortices are identical. Tavantzis and Ting [12] describe the contracting and the expanding self-similar solutions as one-parameter families and discuss their stability. Krishnamurthy and Stremler [8] relate the angles of the triangle, the circulation ratios, the collapse time and the distance travelled before collapse. Since that distance is $\sqrt{1 + 4P^2}$ times the initial distance from the collision point, Propositions 1 and 2 also give the least value of this ratio.

For two rings, Aref [2] derived the reduced equations (15) for arbitrary circulations and analyzed in detail equal and opposite circulations, $x = 1$, which do not satisfy (17). Koiller et al. [7] found the collapsing configurations and their rates; Proposition 2 adds the constant K_n and the minimum over the relative rotation. O’Neil [11] proves that for generic circulations three concentric rings have finitely many relative equilibria and collapse configurations, and Demina and Kudryashov [4] give explicit double-ring configurations formed by two regular polygons with arbitrary circulations. We have not found the minimal values of Propositions 1 and 2 stated in the literature.

The elimination in the proof of Proposition 1 applies to any rational circulation ratio μ and yields a polynomial with rational coefficients satisfied by both branch minima.

Data availability. The verification program and its output are available at <https://github.com/SharpMeow/GENChase> in the directory `research/`.

This work was prepared with AI assistance. The author takes full responsibility for its content.

References

- [1] H. Aref, Motion of three vortices, *Phys. Fluids* **22** (1979) 393–400.
- [2] H. Aref, Point vortex motions with a center of symmetry, *Phys. Fluids* **25** (1982) 2183–2187.
- [3] H. Aref, Self-similar motion of three point vortices, *Phys. Fluids* **22** (2010) 057104.
- [4] M. V. Demina and N. A. Kudryashov, Rotation, collapse, and scattering of point vortices, *Theor. Comput. Fluid Dyn.* **28** (2014) 357–368.
- [5] T. Gotoda, Self-similar motions and related relative equilibria in the N -point vortex system, *J. Dyn. Differ. Equ.* **33** (2021) 1759–1777.
- [6] W. Gröbli, *Specielle Probleme über die Bewegung geradliniger paralleler Wirbelfäden*, Inaugural-Disser-tation, Göttingen; Zürcher und Furrer, Zürich, 1877.
- [7] J. Koiller, S. Pinto de Carvalho, R. Rodrigues da Silva and L. C. Gonçalves de Oliveira, On Aref’s vortex motions with a symmetry center, *Physica D* **16** (1985) 27–61.
- [8] V. S. Krishnamurthy and M. A. Stremler, Finite-time collapse of three point vortices in the plane, *Regul. Chaotic Dyn.* **23** (2018) 530–550.
- [9] X. Leoncini, L. Kuznetsov and G. M. Zaslavsky, Motion of three vortices near collapse, *Phys. Fluids* **12** (2000) 1911–1927.
- [10] E. A. Novikov and Yu. B. Sedov, Vortex collapse, *Sov. Phys. JETP* **50** (1979) 297–301 [*Zh. Eksp. Teor. Fiz.* **77** (1979) 588–597].
- [11] K. A. O’Neil, Relative equilibrium and collapse configurations of heterogeneous vortex triple rings, *Physica D* **236** (2007) 123–130.
- [12] J. Tavantzis and L. Ting, The dynamics of three vortices revisited, *Phys. Fluids* **31** (1988) 1392–1409.